# YILI YANG

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## RESEARCH SCOPE

My research develops/applies advanced AI methodologies to solve complex scientific problems, especially in geoscience, environmental and climate sciences. I create data/AI-driven approaches that accelerate scientific discovery, improve predictions, and transform our understanding of our changing planet through interdisciplinary collaborations.

## EDUCATION

**University of Edinburgh, UK**

**PhD in Petrophysics** Oct 2016 - Apr 2021

**MSc (Research) in Geology, Distinction** August 2015 - August 2016

**BSc Geology** Sep 2011 - May 2015

## EXPERIENCE

**Data Scientist, Woodwell Climate Research Center, MA, USA (remote)** Jan 2022 -

- Project Lead: the pan-Arctic Retrogressive Thaw Slumps (RTS) mapping initiative, applying state-of-the-art deep learning techniques to satellite imagery for detecting and quantifying permafrost thaw features, directly contributing to climate feedback modelling and carbon cycle research
- Designed and built the ARTS scientific dataset, establishing a benchmark training resource for deep learning applications in environmental monitoring
- Developed innovative machine learning workflows for multimodal data integration, combining various satellite imagery types and field observations to enhance feature detection accuracy
- Developed A Zero-shot, interactive segmentation workflow for satellite image labelling using the Segment Anything Model (SAM)
- Collaborated with UCONN in using Location Embeddings to improve satellite image segmentation with vision foundation models.
- Collaborated with ASU in using big geospatial data to produce pan-Arctic statistics.
- Collaborated with UIUC in optimising the workflow of the Deep Learning model inference in big geospatial data (Arctic-scale) in GCP.
- Collaborated with ASU in modelling abrupt thaw initialisation with remote sensing and deep neural networks.
- Collaborated in two wildfire projects using deep learning for wildfire mapping on satellite imagery and using environmental variables to predict wildfire using machine learning
- Collaborated in carbon flux time series imputation using machine learning

- Lead annual machine learning workshop series for Woodwell researchers, providing training in advanced AI techniques for environmental science applications including image processing and time-series forecasting. Designing progressive curriculum materials and adapting teaching methods for diverse learner backgrounds.
- Mentored interns and research assistants, created training materials for interns and research assistants. Mentee Works were presented at the AGU annual conference.

**Data Science Fellow, Faculty.ai, London, UK** Sep 2021 - Dec 2021

- Completed intensive fellowship in Machine Learning and Artificial Intelligence, focusing on developing practical AI solutions for complex data challenges. Include trainings and courses for professional data scientist and an industry placement.

**PhD Researcher, International Centre for Carbonate Reservoirs, Edinburgh, UK** Sep 2017 - Sep 2020

- Funded by PetroBras, petrophysics and digital rock experimental study.
- Conducted doctoral research on multiphase fluid flow in porous media, developing novel computational methods for analyzing complex flow dynamics, funded by Petróleo Brasileiro
- Parcipitated in petrophysical experiments at the Diamond Light Source UK, the Advanced Photon Source Argonne National Laboratory, US and the Swiss Light Source Paul Scherrer Institute, CH

## PUBLICATIONS, CONFERENCES AND DATA SETS

**Yang, Yili**, Arav Agarwal, Marlena Bartkus, Heidi Rodenhizer, Wenwen Li, Hyunho Lee, Chia-Yu Hsu, Greg Fiske, Stefano Potter, Anna Liljedahl, et al. Keeping pace with a changing planet: An interactive segmentation framework for refining delineations of dynamic earth features with the segment anything model. *International Journal of Applied Earth Observation and Geoinformation*, 147:105187, 2026.

Anna-Maria Virkkala... Data-driven modeling of carbon dioxide and methane fluxes across the arctic-boreal region: recent achievements and future opportunities. submitted to Global Change Biology.

Amal S. Perera, David Fernandez, Chandi Witharana, Elias Manos, Michael Pimenta, Anna K. Liljedahl, Ingmar Nitze, **Yang, Yili**, Todd Nicholson, Wenwen Li, and Chia-Yu Hsu. Pan-arctic permafrost landform and human-built infrastructure feature detection with vision transformers and location embeddings, 2026. doi:10.1109/JSTARS.2025.3648673.

Stefano Potter, **Yang, Yili**, Arden Burrell, Anna Talucci, Andrew A Clelland, Sander Veraverbeke, James T Randerson, Scott J Goetz, Logan T Berner, Michelle C Mack, et al. Burned area mapping across the arctic-boreal zone with landsat and sentinel-2 imagery. *International Journal of Remote Sensing*, pages 1–29, 2026.

Andrew A. Clelland, Gareth J. Marshall, Robert Baxter, Stefano Hosking, J.Scottand Potter, **Yang, Yili**, Anna C. Talucci, Joshua M. Rady, Hélène Genet, and Brendan M. Rogers. Machine learning projections of fire occurrence in the arctic-boreal zone for 2025-2100 under different ssp scenarios. In preparation.

Wenwen Li, Chia-Yu Hsu, Sizhe Wang, Zhining Gu, **Yang, Yili**, Brendan M. Rogers, and Anna Liljedahl. A multi-scale vision transformer-based multimodal geoai model for mapping arctic permafrost thaw. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, pages 1–15, 2025. doi: 10.1109/JSTARS.2025.3564310.

**Yang, Yili**, Heidi Rodenhizer, Brendan M Rogers, Jacqueline Dean, Ridhima Singh, Tiffany Windholz, Amanda Poston, Stefano Potter, Scott Zolkos, Greg Fiske, et al. A collaborative and scalable geospatial data set for arctic retrogressive thaw slumps with data standards. *Scientific Data*, 12(1):18, 2025.

**Yang, Y**, H Rodenhizer, and J Dean. Arctic retrogressive thaw slumps (arts): digitisations of pan-arctic retrogressive thaw slumps, 1985-2021. Arctic Data Center, 2024. doi:10.18739/A2PK0738B.

Zhining Gu, Wenwen Li, Chia-Yu Hsu, Sizhe Wang, **Yang, Yili**, Brendan M Rogers, and Anna Liljedahl. A multi-scale vision transformer-based multimodal geoai model for mapping arctic permafrost thaw. *Available at SSRN 4762408*, 2024.

Wenwen Li, Chia-Yu Hsu, Sizhe Wang, Yezhou Yang, Hyunho Lee, Anna Liljedahl, Chandi Witharana, **Yang, Yili**, Brendan M Rogers, Samantha T Arundel, et al. Segment anything model can not segment anything: Assessing ai foundation model's generalizability in permafrost mapping. *Remote Sensing*, 16(5):797, 2024.

Heidi Rodenhizer, **Yang, Yili**, Greg Fiske, Stefano Potter, Tiffany Windholz, Andrew Mullen, Jennifer D Watts, and Brendan M Rogers. A comparison of satellite imagery sources for automated detection of retrogressive thaw slumps. *Remote Sensing*, 16(13):2361, 2024.

**Yang, Yili**, Heidi Rodenhizer, Brendan M Rogers, Jacqueline Dean, Ridhima Singh, Tiffany Windholz, Amanda Poston, Stefano Potter, Scott Zolkos, Greg Fiske, et al. Arts: a scalable data set for arctic retrogressive thaw slumps. In *EGU General Assembly Conference Abstracts*, page 1365, 2024.

Brendan M Rogers, Susan Natali, Robin Bronen, John P Holdren, Patricia Cochran, Kyle Andreas Arndt, Elchin E Jafarov, Melissa Shapiro, Anna Virkalla, **Yang, Yili**, et al. Permafrost pathways: Connecting science, people, and policy to advance understanding of the local to global impacts of permafrost thaw and develop just and equitable responses. In *AGU Fall Meeting Abstracts*, volume 2023, pages B33H–233, 2023.

**Yang, Yili**, Brendan M Rogers, Greg Fiske, Jennifer Watts, Stefano Potter, Tiffany Windholz, Andrew Mullen, Ingmar Nitze, and Susan M Natali. Mapping retrogressive thaw slumps using deep neural networks. *Remote Sensing of Environment*, 288:113495, 2023a.

Stefano Potter, Arden Llewellyn Burrell, Anna Talucci, **Yang, Yili**, Logan T Berner, Scott J Goetz, Sander Veraverbeke, James T Randerson, Susan Natali, and Brendan M Rogers. Mapping alaskan and canadian wildfires using convolutional neural networks. In *AGU Fall Meeting Abstracts*, volume 2023, pages B31M–2252, 2023.

Stefano Potter, **Yang, Yili**, Arden Burrell, Anna Talucci, Sander Veraverbeke, James T Randerson, Scott Goetz, Logan Berner, Michelle Mack, Xanthe Walker, et al. Mapping arctic-boreal burned area in north america using a convolutional neural network with landsat and sentinel-2 imagery. *Available at SSRN 4803815*, 2024.

Heidi Rodenhizer, **Yang, Yili**, Greg Fiske, Stefano Potter, Tiffany Windholz, Andrew Mullen, Jennifer Watts, and Brendan M Rogers. Comparison of three different imagery sources for retrogressive thaw slump detection using a unet3+ deep learning model. In *AGU Fall Meeting Abstracts*, volume 2023, pages C21E–1276, 2023.

Wenwen Li, Chia-Yu Hsu, Sizhe Wang, Yezhou Yang, Anna K Liljedahl, Hyunho Lee, Chandi Witharana, **Yang, Yili**, Brendan M Rogers, Samantha T Arundel, et al. Geoai foundation models for vision: A view from ai augmented environmental mapping. *AGU23*, 2023.

**Yang, Yili**, Brendan M Rogers, Greg Fiske, Jennifer Watts, Stefano Potter, Tiffany Windholz, Andrew Mullen, Ingmar Nitze, and Sue Natali. Mapping retrogressive thaw slumps using satellite data with deep learning. In *EGU General Assembly Conference Abstracts*, pages EGU–1675, 2023b.

Andrew L Mullen, Jennifer D Watts, Brendan M Rogers, Mark L Carroll, Clayton D Elder, Jonas Noomah, Zachary Williams, Jordan A Caraballo-Vega, Allison Bredder, Eliza Rickenbaugh, and **others**. Using high-resolution satellite imagery and deep learning to track dynamic seasonality in small water bodies. *Geophysical Research Letters*, 50(7):e2022GL102327, 2023.

Ridhima Singh, **Yang, Yili**, and Brendan M Rogers. Mapping retrogressive thaw sumps (rts) using transformer-based neural networks. In *AGU Fall Meeting Abstracts*, volume 2022, pages C52C–0362, 2022.

**Yang, Yili**, Sohan Seth, Ian B Butler, and Florian Fusseis. Fast segmentation of 4d microtomography volumes from core-flooding experiments in porous rock using convolutional neural network. *Authorea Preprints*, 2022.

S Marti, F Fusseis, IB Butler, C Schlepütz, F Marone, J Gilgannon, R Kilian, and **Yang, Y**. Time-resolved grain-scale 3d imaging of hydrofracturing in halite layers induced by gypsum dehydration and pore fluid pressure buildup. *Earth and Planetary Science Letters*, 554:116679, 2021.

Sina Marti, Florian Fusseis, Ian B Butler, Christian Schlepütz, Federica Marone Welford, James Gilgannon, Rüdiger Kilian, and **Yang, Yili**. Chemical-mechanical-hydraulic coupling in deforming, dehydrating halite-gypsum rocks-implications for basal detachments in thin-skinned tectonics. In *EGU General Assembly Conference Abstracts*, page 9560, 2020.

**Yang, Yili**, Ian B Butler, Florian Fusseis, Rink van Dijke, Sebastian Geiger, and Xianghui Xiao. Immiscible fluid displacement and trapping during a drainage-imbibition cycle in porous carbonate rock imaged by synchrotron x-ray micro-tomography. In *AGU Fall Meeting Abstracts*, volume 2018, pages H41E–08, 2018.

## RESEARCH FUNDINGS

Funding for Climate Solutions: A Generic Climate AI Framework for Multi-domain Time Series Prediction, Woodwell Climate Research Center, $99,749

Overall Objectives: This project will attempt to develop a generic climate AI framework that will unify and accelerate time-series inferences across Woodwell Climate Research Center's diverse research domains, from Arctic carbon fluxes to tropical forest dynamics.

Pending/ongoing/submitted funding applications with NASA, NSF, Google.org and the Bezos Foundation

## PARTICIPATED RESEARCH INITIATIVES

**Permafrost Discovery Gateways** Funded by Google.org and NSF, the Gateway is making information of permafrost conditions available throughout the Arctic by providing access to big geospatial products and tools to allow exploration and discovery for researchers, educators, and the public at large. This includes providing automated monthly monitoring of permafrost thaw during the snow-free season from satellite imagery.

**Permafrost Pathways** Funded through the TED Audacious Project — a collaborative funding initiative catalyzing big, bold solutions to the world's most urgent challenges. Through a joint effort between

Woodwell Climate Research Center, the Arctic Initiative at Harvard Kennedy School, and the Alaska Institute for Justice, Permafrost Pathways brings together leading experts in climate science, policy action, and environmental justice to inform and develop adaptation and mitigation strategies to address permafrost thaw.

**GeoAI Challenge** Cooperate with the Cyberinfrastructure and Computational Intelligence Lab and Prof. Wenwen Li at Arizona State University to host an NSF-funded AI challenge (Cyber2A Arctic Challenge) using remote sensing imagery to detect permafrost thaw.

## MENTORSHIP

Multi-domain Timeseries Forecasting

Sandeep Poudel 2026
Dogukan Teber 2025

Satellite Image Labelling Workflow and Training Image Generation

Robb Young 2025
Marly Bartkus 2025

Deep Learning label data curation for ESRI-Woodwell Collaboration

Marly Bartkus 2025

Automatic Improvement of Geospatial Digitisation using Segment Anything Model

Arav Agarwal 2025
Milagros Becerra 2025

Estimation of the Volumetric Loss of Retrogressive Thaw Slumps using DEM

Margo Moceyunas 2024

Using Vision-Transformer to segment Retrogressive Thaw Slumps on Remote Sensing Imagery

Ridhima Singh 2023

## ACADEMIC SERVICES, CONFERENCES AND TALKS

- Manuscripts review: 7
- NSF Proposal review: 1
- Tropical Forest Forever Facility (TFFF): Pathways to Impact, London Climate Action Week, UK 2026
- Accelerating Climate Progress with AI: From Science to Action Workshop, The National Academies, US 2024
- Invited talk at the Arctic Carbon Flux Upscaling Workshop, Sweden 2024
- Invited panel talk at the Esri UC Spatial Analytics Summit, US 2024
- Invited talk at the Cyber2A Workshop, US 2024
- European Geophysical Union Annual Conference, EU 2022, 2023, 2024
- Google Geo-for-Good Summit, US/Singapore 2022, 2023, 2025

- Tri-Polar Remote Sensing Conference, China 2023, 2024
- RTSInTrain Workshop, EU 2023
- NASA Arctic-Boreal Vulnerability Experiment workshop, US 2022
- Invited Permafrost Discovery Gateway Webinar 2022
- Invited webinar at the Royal Meteorological Society: Machine Learning for Atmospheric Sciences: Values and Controversies, UK 2022
- American Geophysical Union annual conference, US 2018